3-Day Induction Curriculum

Centre of Excellence (CoE) in Robotics and Artificial Intelligence

Day 1: CoE Orientation, Robotics Basics & Hardware Familiarization

Session 1: CoE Orientation & Upskilling Opportunities

- Overview of the CoE in Robotics & AI
- How students can use the CoE for upskilling, projects, research, competitions, and internships

Session 2: Fundamentals of Robotics

- Introduction to robotics systems
- Applications of robotics
- Functional blocks of a robotic system

Session 3: Innovation & Engineering Mindset

- Design thinking for engineers

Session 4: Robotics Hardware Components

- Processors and microcontrollers
- Control boards
- Power supply and safety
- Input devices (sensors)
- Output devices (motors, LEDs, buzzers)

Day 2: Sensors, Robotics Coding & Rover Development

Session 1: Sensors & Interfacing

- Types of sensors used in robotics
- Interfacing sensors with control boards

Session 2: Robotics Programming

- Basics of embedded/robotics programming
- Reading sensor data
- Controlling motors and outputs

Session 3: Rover Development – Electronics & Control

- Control board configuration
- Motor driver interfacing
- Sensor integration with rover platform
- Power distribution and signal flow

Session 4: Rover Testing & Debugging

- Functional testing of rover
- Debugging hardware and code issues
- Preparing rover for motor control and race

Day 3: Motor Control, Optimization & Robo Race

Session 1: Motor Control Techniques

- DC motors and motor drivers
- Speed and direction control using PWM

Session 2: Rover Optimization & System Integration

- Sensor–motor coordination
- Logic optimization and calibration
- Performance improvement

Session 3: Robo Race Competition

- Rules and track briefing
- Team-based robo race
- Real-time troubleshooting

Session 4: Evaluation & CoE Roadmap

- Performance evaluation and feedback
- Learning reflection
- Introduction to advanced CoE activities in Robotics & AI
- Replace “rover” with any specific platform name you use